

1. Title Page

- **Page Title:** Exploratory Data Analysis Report of FoodHub Order
- **Dataset Name:** foodhub_order.csv
- **Student Name:** Rajesh Rijal
- **Course / Program:** Data Science with Python
- **Instructor Name:** Mr. Nabin Katwal
- **Institution:** Mindrisers Institute of Technology
- **Submission Date:** 22/02/2026

2. Executive Summary

The exploratory data analysis of the FoodHub order dataset shows that customer demand is concentrated among a limited number of restaurants and cuisine types, which generate the majority of orders and sales. Orders are higher on weekends compared to weekdays, indicating stronger demand during those periods. Most customer ratings are favorable, reflecting overall satisfaction, while variations in food preparation and delivery times highlight areas for operational improvement. Overall, the analysis identifies key revenue-driving segments and opportunities to enhance service efficiency and customer experience.

3. Introduction

- **Background and Context:** The FoodHub includes order details such as restaurant, cuisine type, order amount, day, time, delivery time, and ratings. The analysis examines customer ordering patterns, sales performance, and service efficiency.
- **Purpose of the Analysis:** To identify key trends in customer orders, evaluate restaurant and cuisine performance, and assess operational efficiency using preparation and delivery times. It aims to generate insights that support better business decisions and improve overall service performance.
- **Scope of the study:** To evaluate restaurant performance, cuisine popularity, order amounts, ratings, and operational metrics such as preparation and delivery time. The analysis is limited to the information available within the dataset.

4. Problem Definition

- **Business Problem:** Overdependence on a few restaurants and cuisines, inconsistent preparation and delivery times, and fluctuating weekday demand may lead to revenue imbalance, operational inefficiencies, and potential customer dissatisfaction.
- **Key Questions:**
 - What are the most popular cuisine type?
 - Which restaurants receive the highest number of orders?
 - Do weekends generate more revenue than weekdays?
 - How are customer ratings distributed?

5. Data Description

- **Data Source:** The data source is the FoodHub order dataset, which contains customer order records including restaurant name, cuisine type, order amount, day of order, preparation time, delivery time, and customer ratings.
- **Number of Records and variables:** There are 1,898 records (rows) and 9 variables (columns).
- **Description of Key Features:**
 - **Restaurant Name:** Identifies the restaurant fulfilling the order.
 - **Cuisine Type:** Indicates the type of food ordered.
 - **Order Amount:** Shows the total value of each order (revenue per order).
 - **Day of the Week:** Specifies whether the order was placed on a weekday or weekend.
 - **Food Preparation Time:** Time taken by the restaurant to prepare the order.
 - **Delivery Time:** Time taken to deliver the order to the customer.
 - **Rating:** Customer feedback score reflecting satisfaction level.

6. Data Preparation

- **Data Cleaning Steps:**
 - The order_id column was removed as it was not required for analysis.
 - Inconsistent restaurant names were cleaned by removing special characters and standardizing formatting.
 - The rating column was converted to numeric format for proper analysis.
 - Missing rating values were handled by replacing them with the mode.
 - Data was checked for null values and duplicates to ensure data quality.
- **Missing Value Treatment:**
 - Missing values were identified in the rating column.
 - The missing ratings were replaced using the mode (most frequent value).
 - After treatment, the dataset contained no missing values.
- **Outliers Handling:** There are no outliers.
- **Data transformations:** The rating column was converted from object type to integer for further analysis.

7. Exploratory Data Analysis

7.1 Univariate Analysis

- **cuisine_type:** Analyzed using count plots and frequency distribution to identify the most popular cuisines.
- **restaurant_name:** Examined to determine restaurants with the highest number of orders.
- **day_of_the_week:** Analyzed to understand order distribution across weekdays and weekends.
- **cost_of_the_order:** Explored using descriptive statistics and distribution plots to study spending patterns.
- **rating:** Analyzed to evaluate customer satisfaction levels.
- **food_preparation_time:** Examined to understand preparation time distribution.
- **delivery_time:** Analyzed to observe delivery performance patterns.

7.2 Bivariate Analysis

- **cuisine_type & rating:** Analyzed using groupby aggregation to examine how customer ratings vary across different cuisine types, helping identify which cuisines receive higher overall customer satisfaction.
- **day_of_the_week & cost_of_the_order:** Examined to understand differences in total sales between weekdays and weekends, highlighting variations in customer spending behavior based on the day.
- **cuisine_type & cost_of_the_order:** Analyzed to determine which cuisine types generate the highest revenue, identifying key contributors to overall business performance.

7.3 Multivariate Analysis

The `cuisine_type`, `day_of_the_week`, `order_amount`, and `rating` columns are used for multivariate analysis. The figure shows that certain cuisine types contribute significantly higher total sales and receive strong customer ratings, indicating high demand and customer preference. It also reveals that sales are higher on weekends compared to weekdays, highlighting peak demand periods that strongly influence overall business performance.

8. Key Insights

- **Cuisine Preference:** A few cuisine types dominate both orders and ratings, making them key drivers of customer demand.
- **Revenue Contribution:** Certain cuisines generate higher total revenue, indicating strong market value and pricing impact.
- **Sales by Day:** Sales are higher on weekends, showing clear peak demand periods.
- **Customer Satisfaction:** Ratings differ across cuisines, with some consistently performing better in customer satisfaction.
- **Revenue Concentration Trend:** Revenue is concentrated among a small number of cuisines, showing that only a few categories drive most of the total sales.

9. Recommendations

- **Focus on High-Performing Cuisines:** Prioritize top-performing cuisines in marketing and promotions to maximize sales and customer engagement.
- **Optimize Weekend Strategy:** Leverage higher weekend demand through targeted offers and promotions to increase revenue.
- **Improve Low-Rated Cuisines:** Address quality and service issues in lower-rated cuisines to enhance customer satisfaction.
- **Strategic Resource Allocation:** Allocate operational resources toward high-demand cuisines to ensure efficiency during peak periods.

10. Conclusion

The analysis shows that FoodHub's performance is driven by a small number of popular cuisines and restaurants that generate most orders and revenue. Customers typically place moderately priced orders, and demand varies between weekdays and weekends, affecting revenue patterns. Service is generally efficient and ratings are mostly positive, but longer preparation and delivery times reduce customer satisfaction, and missing ratings limit feedback insights. Overall, focusing on top cuisines, improving delivery efficiency, and strengthening feedback collection can significantly enhance business performance.

11. Weakness in the Project

- **Biased handling of missing ratings:** Missing rating values were filled using the mode, which can distort customer satisfaction patterns and hide real feedback gaps.
- **Limited interpretation of visualizations:** Charts are created, but clear explanations and business implications are often missing, making conclusions less actionable.